Ieva Bagdonaviciute

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EDUCATION

Stanford University, School of Engineering Expected June 2028
Master of Science in Computer Science, Artificial Intelligence Specialization

University of Illinois at Urbana – Champaign, Grainger College of Engineering May 2026
Bachelor of Science in Computer Engineering, Highest Honors | Undergraduate Researcher Certificate GPA: 3.90/4.0

EXPERIENCE

Stanford University | NeuroAILab Sep 2026 – present
Research Assistant Stanford, CA, USA

- Lead **real-robot evaluation** of bimanual manipulation policies by designing a standardized YAM benchmark that quantifies pick-and-place accuracy and stability across diverse objects.
- Present **PsiBorg at CoRL 2026** in a live demo of our 8B-parameter robot foundation model, which uses goal images to guide actions, predicts future observations, and estimates time to goal.
- Prepared **2,722 teleoperation clips** for quality review, defining criteria for pickup, transport, and placement and flagging ambiguous clips and recovery sequences.

University of Illinois at Urbana-Champaign | Computer Vision Group Aug 2025 – Aug 2026
Student Researcher Urbana, IL, USA

- Fine-tuned **vision-language models with LoRA** for physical reasoning, investigating dataset scaling, difficulty-aware sample weighting, multimodal embeddings, and prompting strategies to improve model performance.
- Built an end-to-end **RGB-D data and generative reconstruction pipeline**, integrating monocular depth estimation, physics-based simulation, generative models, and automated VLM feedback across multiple datasets and conditions.
- Designed quantitative experiments using **PSNR, SSIM, LPIPS, CIEDE2000, and mixed-effects statistical models** to evaluate model performance and analyze reconstruction quality; contributed to a preprint

Microsoft Research | AI Frontiers Laboratory May 2025 – Aug 2025
Research Intern Redmond, WA, USA

- Built an **end-to-end VLM evaluation pipeline** in Microsoft's open-source EUREKA framework, **preprocessing 10,000+ video samples for large-scale evaluation** and benchmarking **6 frontier models** on physical reasoning.
- Implemented an **LLM-as-a-Judge** system and 6 custom test types to evaluate perception, reasoning, and model failure modes across 3 physical reasoning datasets.
- Developed confidence–accuracy and perception–reasoning analyses with automated evaluation and visualization; co-authored a preprint and presented at the Microsoft Research Intern Poster Session.

University of Illinois Urbana-Champaign | Prof. Yoram Bresler's Laboratory Aug 2024 – May 2025
Research Assistant Urbana, IL, USA

- Developed **Python pipelines for 3D ultrasound/EIT simulation and U-Net medical-image segmentation**, integrating FOCUS, k-Wave, and AbdomenAtlas.
- Generated **20,000+ multimodal parameter maps** from **19 segmented abdominal organs and body-composition tissues** for future multimodal model training.

The Swiss Federal Institute of Technology (EPFL) | Computational Quantum Science Laboratory May 2024 – Aug 2024
Machine Learning Research Intern Lausanne, Switzerland

- **Developed a differentiable MCMC framework in JAX**, implementing Metropolis-Hastings sampling, gradient estimation, and optimization to model the 2D Ising phase transition.
- **Benchmarked 2 gradient-estimation methods** across increasing system sizes, achieving **>96% phase-transition prediction accuracy** on larger systems and characterizing scalability tradeoffs.
- Presented a research report and a poster at the Engineering Research Fair and International Research Symposium

Teltonika IoT Group May 2023 – Jul 2023
Embedded Software Engineering Intern Vilnius, Lithuania

- Operated with STM32CubeIDE software to configure a functional STM32F413 microcontroller on the FMB640 hardware platform, adhering to Telematics coding style
- Developed software layers using CMSIS-RTOS API v2 and STM32 low layer library, encompassing drivers (GPIO and UART), APIs (UART, command, debug, LED, and heap), and Application (CLI) tailored for the tracker's optimal performance in reading messages and performing the given commands
- Conducted and delivered weekly research presentations on topics such as UART, RS-485, MQTT, I2C, I2S, CRC, Ethernet and more, showcasing strong analytical and communication skills

PUBLICATIONS & PRESENTATIONS

- **PsiBorg.** Live robotics demo with Stanford NeuroAILab at the Conference on Robot Learning (CoRL), Austin, TX, 2026. https://neuroailab.github.io/psi-website/psiborg/
- *Aghajanzadeh, S., Wang, Y.,* ***Bagdonaviciute, I.****, & Forsyth, D.* (2026). *VLMs Win: A Systematic Evaluation of Underwater Image Reconstruction.* Preprint, arXiv:2608.11425v1 [cs.CV]. https://arxiv.org/abs/2608.11425
- ***Bagdonaviciute, I.*** *& Vibhav Vineet* (2025). *Physics Knowledge in Frontier Models: A Diagnostic Study of Failure Modes.* Preprint arXiv:2510.06251v2 [cs.CV]. https://arxiv.org/abs/2510.06251
- ***Bagdonaviciute, I.*** *& Vibhav Vineet* (2025). *Does Physics Knowledge Emerge in Frontier Models?* Poster presented at the Microsoft Research Intern Poster Session, Redmond, WA.
- ***Bagdonaviciute, I.*** (2024). *Optimization of the Ising Model: Comparing Gradient Estimation Methods.* Poster presented at the Engineering Research Fair and International Research Symposium, University of Illinois Urbana-Champaign.
- ***Bagdonaviciute, I.*** *& Gavin N. R.* (2024). *Analyzing Industrial Software via Open-Source Frameworks*. Poster presented at the Undergraduate Research Symposium, University of Illinois Urbana-Champaign.

SELECTED PROJECTS

F1TENTH Autonomous Vehicle | UIUC ECE 484 Principles of Safe Autonomy | ROS2, OpenCV, RealSense | 2025

- Developed a ROS2 line-following node using camera-based centerline detection, PD steering, and configurable speed control.
- Integrated RealSense depth sensing for obstacle stopping; debugged camera I/O and ROS2 launch issues using image topics and frame rates.

UR3 Robotic Manipulation | UIUC ECE 470 Introduction to Robotics | ROS, OpenCV | 2026

- Built an image-to-robot drawing pipeline combining OpenCV contour extraction, coordinate transformations, and inverse kinematics to reproduce images with a UR3-mounted marker.
- Programmed Tower of Hanoi pick-and-place sequences using ROS commands and suction-gripper feedback.

Women in Electrical and Computer Engineering | UIUC | 2022–2026

- **Flower Power:** Led a 4-person sub-team building a solar-tracking robot with photoresistors, servo motors, and mobile app override; earned two Engineering Open House awards.
- **Robot S.U.N.D.A.Y.:** Built a six-DOF robotic arm with a team, integrating a Flutter app, HC-05 Bluetooth, Arduino UNO, and six servo motors.

Secret Garden | 1st Place - Collaborative Agent Hackathon at Stanford | 2026

- Built a privacy-preserving multi-agent event planner using Flower, enabling decentralized AI agents to negotiate shared plans while keeping users' calendars, budgets, and preferences local.

Google Developers Student Club | Founder and Lead | 2023–2024

- Collaborated with Google to organize various social events focusing on technology and career development
- Supervised three committees, including software development teams building student-focused solutions such as the carpooling app UNIride and volunteering platform DeedMatch
- Led and built a community of 35+ like-minded and technology-driven people and promoted inclusivity and collaboration by creating an open environment for students from diverse technical and cultural backgrounds

AWARDS & HONORS

- 1st Place Winner, Stanford University Flower.ai Hackathon (2026) – awarded first place for developing a privacy-preserving agentic AI system.
- Undergraduate Researcher Certificate (2026) – awarded in recognition of sustained and significant undergraduate research contributions
- Illinois Engineering Achievement Scholar (2024, 2025) – awarded to top engineering students at UIUC
- Illini Moms Association Scholar (2025) – awarded to 7 UIUC students for academic excellence, leadership, and service
- Excellence Research Scholar (2024) – awarded during research internship at EPFL, Switzerland
- Global Research Scholar (2024) – selected for academic excellence and international research experience
- Phi Eta Sigma Honor Society (2023) – recognized for academic excellence